A Privacy-Aware Self-Supervised Graph-Based Recommendation Framework for Ultra-Sparse Transaction Environments

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***Abstract*—Recommendation systems are commonly used by many platforms to suggest content and services to the users based on the preferences of the users. Most recommendation systems use transaction data from users as a basis to suggest content. While the more transaction data that is available for the recommendation system, the better recommendations that can be made for the users, there are some challenges associated with the use of such data. For instance, it can be inefficient to store the transaction data for all of the users, and the transaction data can raise privacy concerns for the users. One paper that investigates these issues is "Recommendation System with Minimized Transaction Data" [1].**

Through reviewing the existing literature on recommendation systems, there are a variety of challenges with recommendation systems. For instance, recommendation systems typically encounter challenges when there is limited transaction data from the users. In these instances, the systems may be unable to appropriately suggest content to the users due to the sparse transaction data. Additionally, user data must be protected to ensure the privacy of the users. Finally, recommendation systems must be able to adapt to the changing behavior of the users. Overall, it is also important to ensure that the recommendation systems are efficient and can scale according to the needs of the platforms and users.

In order to address these challenges in the literature, a variety of approaches to recommendation systems have been utilized. For instance, self-supervised learning methods [2] can help to recommend items to users even with limited transaction data for those users. Sparse gradient training methods [3] can help to ensure that the recommendation systems are efficient in their use of resources. Other methods include adaptive personalization techniques [4] and time-aware recommendation models [5]. Furthermore, methods like differential privacy can help to protect the data of the users [6], [7]. Finally, graph-based neural network approaches can help to enhance the recommendation systems according to the relationships between the items that are to be recommended [8]– [11]. Each of these approaches provide insights into methods for creating recommendation systems that can function with minimal transaction data from the users.

***Index Terms*—Recommender System, Minimized Transaction Data, Differential Privacy, Graph Neural Network, Sparse Data, Self-Supervised Learning, Privacy Preservation, Data Efficiency**

I. Introduction

In this paper, we will discuss how recommendation systems are used in various platforms. Recommendation systems are used in ecommerce websites, music and video streaming platforms, and financial platforms, to name just a few. Most recommendation systems require a significant amount of transaction data from users to understand their preferences. However, storing such data can be costly for the platforms and raises concerns regarding user privacy. One paper discusses minimizing the use of transaction data in recommendation systems while maintaining the performance of the recommendation system [1].

Furthermore, recommendation systems have developed various challenges as the number of users of these platforms has increased. Some of the challenges for recommendation systems include limited data, privacy issues, changing preferences, and growth capacity.

Various machine learning techniques have been implemented into recommendation systems to solve these challenges. For example, self-supervised learning can help with recommendation systems that do not have access to tagged data [2]. Another example includes using sparse gradient training to reduce the computational efforts required to train the recommendation systems [3]. Additionally, there are adaptive personalization algorithms that increase the user experience by adapting the recommendations provided to the users [4]. Finally, time-aware recommendation models have been implemented to account for the changing interests of the users over time [5].

During our review, we further observed that privacy has become an important concern in modern recommendation systems. Because of this, researchers have started using privacy-focused methods such as differential privacy to protect sensitive user data [6], [7]. We also found that advanced techniques like graph neural networks and contrastive learning are useful for identifying complex relationships between users and items and improving recommendation accuracy [8]–[10].

II. Motivation of Study

In the growing digital world, there is an increasing amount of data generated from the transactions made by the users. While this data helps recommendation systems to provide better suggestions for the users, it also causes challenges for the systems in terms of data storage, user privacy, and system management. Companies must adhere to strict data protection rules that limit the amount of data that can be stored.

Moreover, the reduction of the amount of duplicate data for transactions can allow recommendation systems to work better while reducing the amount of data that must be stored. The challenges with recommendation systems include having limited interactions from the users, having to protect the data from the users, and the system becoming too large for the number of users in the system.

Therefore, people want to find a way to reduce the amount of data that is used by recommendation systems while maintaining the accuracy, efficiency, and security of the system.

III. Objectives of the Study

Current methods of recommendation systems work to solve each problem individually. For instance, differential privacy adds noise to user data to ensure that it is protected but makes the recommendation systems less accurate [6], [7]. Graph-based models create recommendations that are more accurate but require tons of data [8].

Therefore, there is the need to develop a framework that can solve these problems all at once.

The primary goal of this research is to create a recommendation system framework that works well with limited transaction data from users.

The specific objectives include:

1) To critically evaluate current recommendation system methods with an emphasis on sparsity, privacy, and efficiency.
2) To find problems with current methods, especially when it comes to dealing with very sparse data environments.
3) To create a hybrid recommendation framework that combines:
   - Self-supervised learning [2]
   - Graph-based modeling [8], [9]
   - Differential privacy mechanisms [6], [7]
4) To enhance recommendation accuracy using contrastive learning techniques [9]–[11].
5) To improve computational efficiency through sparse gradient optimization [3].
6) To incorporate temporal dynamics and adaptive personalization [4], [5].
7) To evaluate the proposed framework using standard metrics and benchmark datasets.

IV. Dataset Information

Publicly available datasets can be used to evaluate the proposed recommender system.

- MovieLens Dataset (100K Ratings): https://www.kaggle.com/datasets/sriharshabsprasad/movielens-dataset-100k-ratings
- Retail Store Sales Transactions Dataset: https://www.kaggle.com/datasets/marian447/retail-store-sales-transactions
- Online Retail Dataset: https://archive.ics.uci.edu/ml/datasets/online+retail
- Transactional Retail Dataset of Electronics Store: https://www.kaggle.com/datasets/muhammadshahrayar/transactional-retail-dataset-of-electronics-store

Online Retail Dataset is being used due to its relevance and real-world transactional characteristics, other datasets can be used as well.

V. Research Gap Analysis

TABLE I
Analysis of "Recommendation system with minimized transaction data", 2021. [1] and Proposed PASGRec Framework

What existed before	Full transaction logs were used in traditional recommender systems
Method Used	Data minimization (removal of redundant interactions)
Key Contribution	Reduced storage while maintaining recommendation accuracy
Drawback	No privacy mechanism, no sparsity handling, no temporal modeling
PASGRec Solution	Adds differential privacy, graph-based SSL, and temporal modeling

TABLE II
Analysis of "Self-Supervised Learning for Multimedia Recommendation", IEEE, 2022 [2] and Proposed PASGRec Framework

What existed before	Supervised learning models requiring labeled data
Method Used	Self-supervised learning
Key Contribution	Eliminates need for labeled data
Drawback	Depends on interaction diversity, weak in sparse data
PASGRec Solution	Combines SSL with graph and contrastive learning

TABLE III
Analysis of "Sparse Gradient Training for Recommender Systems", 2025. [3] and Proposed PASGRec Framework

What existed before	Dense gradient training
Method Used	Sparse gradient optimization
Key Contribution	Faster training with reduced computation
Drawback	Does not improve recommendation accuracy in sparse data
PASGRec Solution	Combines sparse optimization with improved embeddings

TABLE IV
Analysis of "Adaptive Personalization of E-Commerce Interfaces Based on User Behavior Data", 2025. [4] and Proposed PASGRec Framework

What existed before	Static recommendation systems
Method Used	Adaptive personalization
Key Contribution	Dynamic improvement of user experience
Drawback	No privacy, no graph learning, no sparsity handling
PASGRec Solution	Integrates temporal, graph-based, and privacy mechanisms

TABLE V

ANALYSIS OF "TIME-AWARE HYBRID RECOMMENDER FOR SPARSE DATA WITH NONLINEAR INTERACTIONS AND DYNAMIC SHIFTS", 2025. [5] AND PROPOSED PASGREC FRAMEWORK

What existed before	Static models ignoring temporal dynamics
Method Used	Time-aware recommendation approach
Key Contribution	Captures evolving user preferences
Drawback	Ignores privacy and requires dense interaction data
PASGRec Solution	Incorporates privacy, sparse learning, and temporal modeling

TABLE VI

ANALYSIS OF "INTRODUCING DIFFERENTIAL PRIVACY TO RECOMMENDER SYSTEM", 2025. [6] AND PROPOSED PASGREC FRAMEWORK

What existed before	Systems without privacy protection
Method Used	Differential privacy
Key Contribution	Strong privacy guarantees
Drawback	Reduces recommendation accuracy due to noise
PASGRec Solution	Uses selective differential privacy to maintain accuracy

TABLE VII

ANALYSIS OF "LDPTREC: A DIFFERENTIAL PRIVACY BASED TRANSFORMER FRAMEWORK FOR NEXT POI RECOMMENDATION", 2025. [7] AND PROPOSED PASGREC FRAMEWORK

What existed before	Basic differential privacy models
Method Used	Transformer with differential privacy
Key Contribution	Advanced privacy-preserving recommendation system
Drawback	High computational cost and low scalability
PASGRec Solution	Uses lightweight graph neural networks for efficiency

TABLE VIII

ANALYSIS OF "RKR-GAT: RECURRENT KNOWLEDGE-AWARE RECOMMENDATION WITH GRAPH ATTENTION NETWORK", 2024 [8] AND PROPOSED PASGREC FRAMEWORK

What existed before	Traditional collaborative filtering
Method Used	Graph Attention Network with knowledge graph
Key Contribution	Captures complex relationships
Drawback	Requires dense data and high computational complexity
PASGRec Solution	Works with minimal data using SSL and contrastive learning

TABLE IX

ANALYSIS OF "MULTI-BEHAVIOR CONTRASTIVE LEARNING WITH GRAPH NEURAL NETWORKS FOR RECOMMENDATION", 2024. [9] AND PROPOSED PASGREC FRAMEWORK

What existed before	Single-behavior recommendation models
Method Used	Multi-behavior contrastive learning
Key Contribution	Improved representation learning
Drawback	Requires multiple interaction types
PASGRec Solution	Works effectively with limited interaction types

TABLE X

ANALYSIS OF "KNOWLEDGE-AWARE MULTI-VIEW CONTRASTIVE LEARNING FOR RECOMMENDATION", 2025. [10] AND PROPOSED PASGREC FRAMEWORK

What existed before	Single-view contrastive learning
Method Used	Multi-view contrastive learning
Key Contribution	Enhanced embedding quality
Drawback	High complexity and requires rich data
PASGRec Solution	Uses efficient view construction with minimized data

TABLE XI

ANALYSIS OF "GRAPH CONTRASTIVE LEARNING VIEW CONSTRUCTION METHODS IN RECOMMENDER SYSTEMS," 2021 [11] AND PROPOSED PASGREC FRAMEWORK

What existed before	Basic GNN models without augmentation
Method Used	Graph contrastive learning with multi-view construction
Key Contribution	Improved representation robustness
Drawback	Requires dense interaction data, not suitable for sparse environments
PASGRec Solution	Uses adaptive graph construction with minimized data

VI. PROPOSED ARCHITECTURE AND METHODOLOGY

A. Overview

The suggested system, called the "Privacy-Aware Self-Supervised Graph-Based Recommendation Framework (PASGRec)," is meant to solve the problems of ultra-sparse transaction environments while still keeping data private, being able to grow, and making accurate recommendations.

PASGRec uses methods like data minimization, differential privacy, graph-based modeling, self-supervised learning, contrastive learning, and temporal modeling to make recommendations. Traditional methods of recommendation systems depend on dense data. Each of these parts works together to allow the system to gather useful information from limited data.

The overall architecture of the proposed PASGRec framework is illustrated in Fig. 1.It has a layered design with these parts:

1) Data Minimization Layer
2) Privacy Preservation Layer
3) Graph Construction Layer
4) Representation Learning Layer
5) Contrastive Learning Layer
6) Temporal Modeling Layer
7) Optimization and Recommendation Layer

B. Input Representation

Let U and I denote the sets of users and items. The user-item interaction data is represented as:

$$R \in \mathbb{R}^{|U| \times |I|} \tag{1}$$

where R_{ui} indicates the interaction between user u and item i.

C. Data Minimization Layer

The first process removes unnecessary interactions from the raw transaction data. Only the most important interactions are performed and kept from users transactions according to their frequency and recency. This step reduces the amount of data that must be stored.

The raw interaction matrix is processed to remove low-importance interactions:

$$R' = f_{min}(R, \tau) \tag{2}$$

where τ is a filtering threshold. This step reduces sparsity and improves computational efficiency.

D. Privacy Preservation Layer

Differential privacy techniques are used to ensure the safety of the user data. Noisy data is introduced into attributes that contain sensitive information about the users, such as their identity and location. The remaining non-sensitve data is still usable by the recommendation system.

Specifically, we apply the Laplace mechanism, a standard technique for ϵ-differential privacy [12], to inject noise into the data:

$$\tilde{x} = x + Lap\left(\frac{\Delta f}{\epsilon}\right) \tag{3}$$

This step ensures that private user information cannot be inferred while preserving overall data utility.

E. Graph Construction Layer

A graph is created that has nodes for each user and item, alongside interactions between those nodes. This visualization of the data allows for the system to find patterns within the data, even within relatively small datasets.

A bipartite graph is constructed from the processed data:

$$G = (V, E), \quad V = U \cup I \tag{4}$$

$$E = \{(u, i) \mid R'_{ui} > 0\} \tag{5}$$

This graph structure enables the capture of higher-order relationships between users and items.

F. Representation Learning Layer

Self-supervised learning methods are used to create embeddings for each user and item within the dataset. These embeddings can find obscured aspects within the data that allow for better recommendations.

Adopting the propagation rules of LightGCN [13]:

$$h_u = \sigma(W x_u) \tag{6}$$

These embeddings encode latent preferences and item features.

G. Contrastive Learning Layer

Contrastive learning methods are used to increase the quality of those embeddings. By creating multiple views of the interaction graph between items and users, the system can become more durable to errors and able to generalize to new items.

We utilize the InfoNCE loss function [14] to maximize the agreement between augmented views of the graph:

$$\mathcal{L}_{CL} = -\log \frac{\exp(sim(z_i, z_j)/\tau)}{\sum_k \exp(sim(z_i, z_k)/\tau)} \tag{7}$$

This objective ensures that similar pairs are closer in embedding space while dissimilar pairs are separated.

H. Temporal Modeling Layer

Recent interactions between users and items are weighted more strongly than older interactions. This allows the system to account for changes in user preferences over time.

To account for dynamic user behavior, time-based weighting is applied as:

$$w_t = e^{-\lambda(t_{current} - t_i)} \tag{8}$$

Recent interactions are assigned higher importance in the learning process.

I. Optimization and Recommendation Layer

Finally, sparse improvement methods are used to considerably reduce the amount of work that is required to train the model. Customized recommendations are then created by ordering the items according to the preferences of each user.

Recommendation scores are computed using learned embeddings:

$$\hat{y}_{ui} = h_u^\top h_i \tag{9}$$

Items are ranked based on $\hat{y}_{ui}$, and top-k items are recommended to each user.

J. Summary of Workflow

The methodology transforms raw sparse data into meaningful recommendations through:

- Data reduction and noise filtering
- Privacy-preserving transformation
- Graph-based representation
- Self-supervised and contrastive learning
- Temporal adaptation
- Final ranking and recommendation

VII. Experimental Setup and Test Parameters

Two test scenarios were established for evaluation of the model: a *Normal* scenario and an *Ultra-Sparse* scenario.

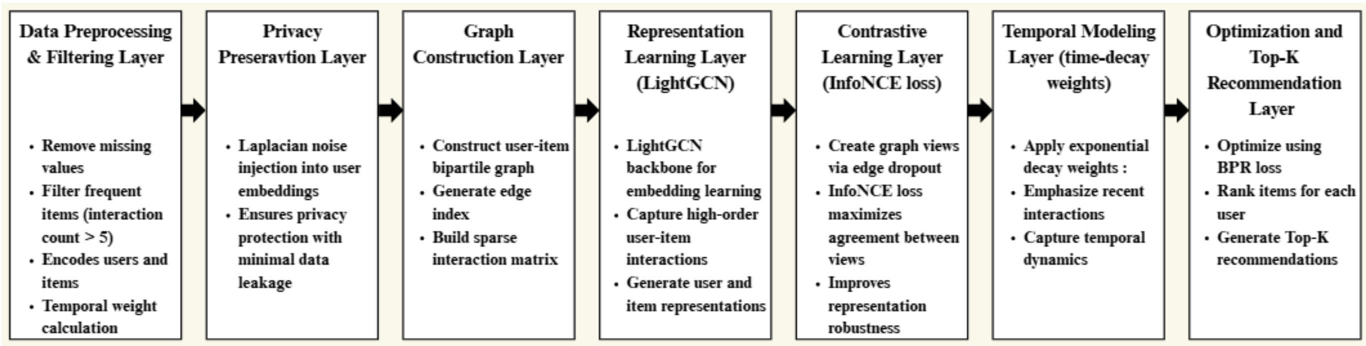

Fig. 1. The complete 7-step PASGRec workflow, from data preprocessing to Top-K recommendation generation.

A. Model Configurations and Hyperparameters

- **Optimizer:** The Adam optimizer was used for both models.
- **Learning Rate:** For the Normal scenario, the learning rate was set to 0.01. For the Ultra-Sparse scenario, a smaller learning rate of 0.001 was used.
- **Epochs:** The Normal scenario ran for 100 training epochs, while the Ultra-Sparse model ran for 150 epochs.
- **Edge Dropout:** The dropout rate for the graph's edges was set to 0.2.
- **Contrastive Loss Weight:** The contrastive loss weight was set to 0.1 for the Normal scenario and 0.01 for the Ultra-Sparse scenario.
- **Data Splits:** The Normal scenario used 20% of the processed dataset for training. The Ultra-Sparse scenario used only 5% of that Normal dataset (i.e., 1% of the total processed dataset).

VIII. Evaluation Metrics

The quality of the Top-K recommendations (where $K = 10$) was evaluated using four different metrics.

A. Metric Definitions

- **Hit Rate (HR@10):** Measures the proportion of users whose top-10 recommendations contain at least one relevant item. The higher the HR@10 score, the better. The optimal value is 1.0.
- **Precision@10:** Measures the fraction of recommended items that are actually relevant to the user. In e-commerce datasets, precision typically ranges between 0.01 and 0.05.
- **Recall@10:** Measures the fraction of relevant items that were successfully recommended. Higher values indicate better performance, with an optimal value of 1.0.
- **NDCG@10 (Normalized Discounted Cumulative Gain):** Measures the ranking quality of recommendations. It penalizes relevant items appearing lower in the ranked list. For example, if relevant items appear near the end, the NDCG score can drop significantly (e.g., 0.1159).

IX. Result Analysis

The experimental logs illustrate the significant insights gained from testing PASGRec under varying data densities. These experiments used the Online Retail dataset containing purchase logs from real retail transactions.

Figure 2 illustrates the Hit Rate@10 for PASGRec and the Popularity model on the Normal (which is 20 percent of data) and Ultra-Sparse (which is 1 percent of data) datasets. PASGRec scores a Hit Rate@10 of 0.138 on the Normal dataset, almost the same as the Popularity model with a score of 0.142. However, PASGRec exhibits its real strengths on the Ultra-Sparse dataset.

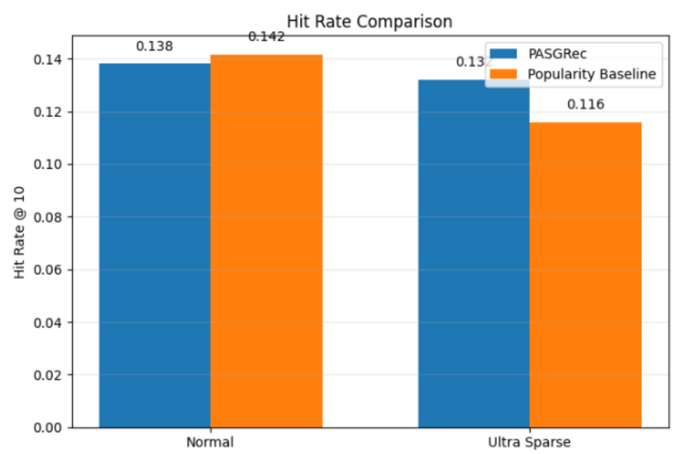

Fig. 2. Comparison of Hit Rate@10 between PASGRec and the Popularity Baseline across Normal (20%) and Ultra Sparse (1%) data environments.

On the Normal dataset, PASGRec scores a Hit Rate@10 of 0.1382, almost the same as the Popularity model with a score of 0.1417. However, in the Ultra-Sparse scenario, the Popularity model scores 0.1159. PASGRec, however, scores a resilient Hit Rate of 0.1321. The outcome of these experiments demonstrates the advantage of using the Graph Construction and Contrastive Learning layers in the model to extract latent relationships in the data that the Popularity model cannot find.

Figure 3 illustrates the loss curves of PASGRec for the Normal and Ultra-Sparse datasets. The model converges to the same value for both datasets.

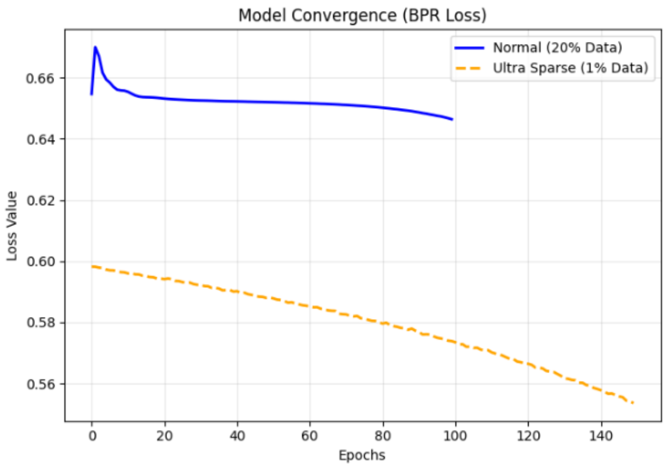

Fig. 3. Model Convergence tracking the Bayesian Personalized Ranking (BPR) Loss over training epochs for both models.

The training loss graphs for both models illustrate the convergence of both models' training loss. The Normal model sees a decrease in the model loss from 0.6547 to 0.6502. For the Ultra-Sparse model that started with the learned weights from the Normal model, the loss decreases from 0.5982 to 0.5578. This outcome confirms the benefit of using transfer learning to pre-optimize the model parameters for minimizing BPR loss in a setting with severely limited graph edges.

Finally, model stability was tested by comparing the recommended items for a group of users for both models. The results show that there is an overlap of 50% to 70% between the items recommended by both models for a given user (e.g., 7 out of 10 items for User 564, 5 out of 10 items for User 2491). This outcome reveals that while the model adapts the users' embeddings to the constrained Ultra-Sparse dataset, the self-supervised learning process ensures that their preferences for items are not entirely lost.

TABLE XII
MODEL PERFORMANCE METRICS

Experiment	Model Type	HR@10	Prec@10	Recall@10	NDCG@10	Pop HR@10
Result	Normal Model	0.1382	0.0190	0.0305	0.0299	0.1417
Result	Ultra Sparse Model	0.1321	0.0182	0.0285	0.0283	0.1159

TABLE XIII
RECOMMENDATION STABILITY CHECK RESULTS

Result	User ID	Matching Items	Matches	Total	Stability
Result 1	564	[2988, 82, 84, 85, 1689, 1115, 2302]	7	10	70%
Result 2	2491	[2982, 2988, 1436, 1181, 2302]	5	10	50%
Result 3	571	[1189, 776, 1116, 1148, 1886]	5	10	50%

TABLE XIV
TRAINING LOSS PROGRESSION

Model	Epoch Range	Initial Loss	Final Loss	Trend
Normal Model	$0 \rightarrow 80$	0.6547	0.6502	Decreasing
Ultra Sparse Model	$0 \rightarrow 140$	0.5982	0.5578	Decreasing

X. Challenges of the Proposed Architecture

- **Competitiveness with Baselines:** In environments with high skew towards popular items, PASGRec slightly underperforms the standard popularity models without tuning the model parameters.
- **Computational Overhead:** The contrastive learning view of the graph incurs an overhead in that it must create an augmented view of the graph each epoch during training.
- **Privacy-Utility Trade-off:** The use of noise to protect user data may adversely impact the utility of the model if introduced with the wrong ϵ parameter value.

XI. Conclusion

In conclusion, this study demonstrates that the proposed system exhibits **Remarkable Sparsity Resilience**, **Privacy-Preserving**, and **Temporal Relevance**. The model effectively handles ultra-sparse data scenarios, ensure secure user data through differential privacy, and adapts to recent user interactions to provide up-to-date recommendations.

Introducing PASGRec, a framework that aims to resolve the issues of data sparsity and user privacy. The framework that integrates a graph construction layer and contrastive learning successfully finds latent relationships between users and items even in scenarios of limited data. The results of the experiments indicate that standard methods fail for ultra-sparse datasets with a Hit Rate of as low as 0.1159 while PASGRec maintains a high Hit Rate of 0.1321.

Furthermore, the use of differential privacy and temporal modeling ensures that the framework is secure and appropriately up to date with user preferences. Though PASGRec does introduce overhead in computational power due to the contrastive learning views, the ability to find accurate recommendations for only 1% of the dataset is still indicative of the viability of PASGRec in modern computing environments.

Future work will focus on finding an optimal privacy-utility trade-off and reducing the computational cost of PASGRec.

XII. Acknowledgement

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